Project Design Phase-II

Data Flow Diagram & User Stories

Date	02 October 2026
Team ID	SWTID-2026-9980
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Data Flow Diagram:

A Data Flow Diagram (DFD) shows how information moves through the Script-Controlled ACL system. It shows the user request, ServiceNow application, ACL script evaluation, and the final access decision for Institution Details records based on the required field value and user role.

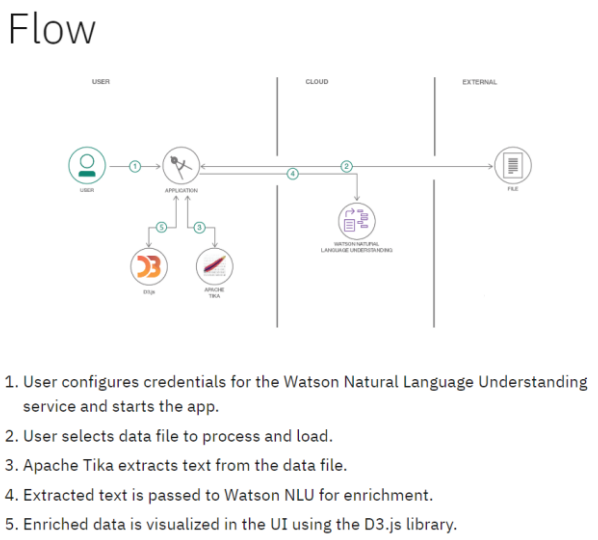
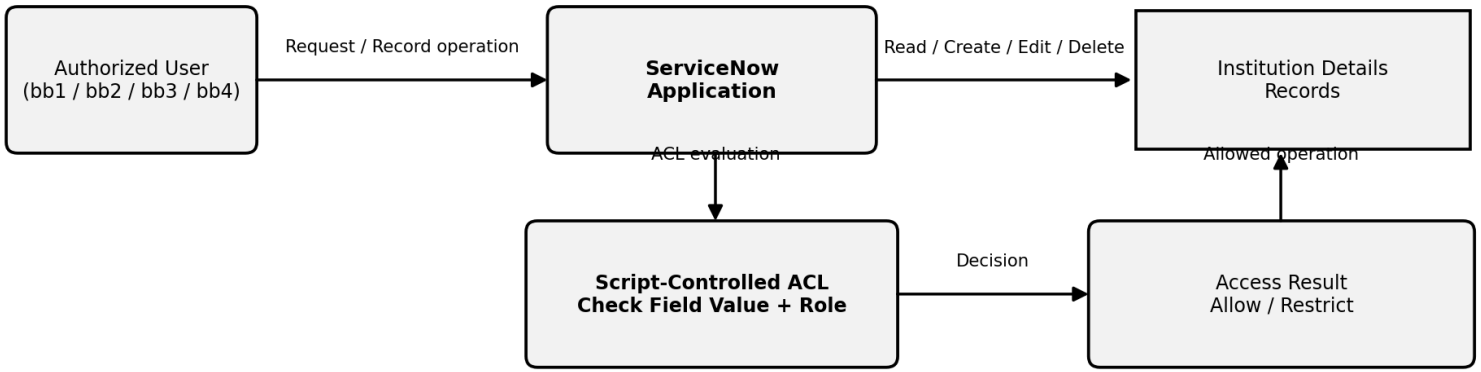


Figure 1: Level 0 Data Flow Diagram for Script-Controlled ACL

DFD Level 0 – Script-Controlled ACL

Example field rule: access is controlled according to the Branch field (e.g., EEE) and assigned roles.



User Stories

The following user stories define the functional requirements for implementing field-value-based record access using Script-Controlled ACLs in ServiceNow.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
ServiceNow Administrator	Create Data Structure	USN-1	As an administrator, I can create the Institution Details table with the required branch and record fields.	The table is created and the required fields are available for ACL evaluation.	High	Sprint-1
ServiceNow Administrator	Define Roles	USN-2	As an administrator, I can assign the required roles (bb1, bb2, bb3 and bb4) to users.	Users receive the correct role and can be identified by the ACL during access evaluation.	High	Sprint-1
Authorized User	Record Access	USN-3	As an authorized user, I can view Institution Details records only when the field value satisfies the access rule.	Records matching the configured field-value condition are visible; restricted records are not accessible.	High	Sprint-1
Authorized User	Record Operations	USN-4	As an authorized user, I can create and edit permitted Institution Details records according to the ACL rule.	Create and update operations are allowed only when the ACL condition and required role are satisfied.	High	Sprint-2
Authorized User	Delete Control	USN-5	As an authorized user with the required role, I can delete permitted records when the field-value rule allows deletion.	Delete is successful for permitted records and denied for restricted records.	High	Sprint-2
ServiceNow Administrator	Script-Controlled ACL	USN-6	As an administrator, I can use a script-controlled ACL to evaluate a record field value before granting access.	The ACL script checks the configured field value and returns the correct allow or restrict decision.	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
ServiceNow Administrator	ACL Verification	USN-7	As an administrator, I can test the ACL with users having different roles and field values.	Users with the required roles can perform permitted operations, while other users are restricted.	High	Sprint-3
ServiceNow Administrator	Security Validation	USN-8	As an administrator, I can verify that unauthorized users cannot bypass the record-level access restriction.	Unauthorized access is denied and permitted users retain the expected record operations.	Medium	Sprint-3

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